

# Piyusha Akash

*Offensive Security Engineer*

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[github.com/0x3xp](https://github.com/0x3xp) | [tryhackme.com/p/0x3xp](https://tryhackme.com/p/0x3xp) | [hackthebox.com/users/1981311](https://hackthebox.com/users/1981311)  
[crackmes.one/user/0x3xp](https://crackmes.one/user/0x3xp) | [youtube.com/@InfoSecLK](https://youtube.com/@InfoSecLK)

## SUMMARY

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Offensive Security Engineer and low-level security researcher specializing in Windows internals, reverse engineering, exploit development, penetration testing, web application security assessment, and custom tooling. Ranked in the Top 5% globally on TryHackMe, securing 1st place in the Silver and Bronze leagues, 11th place in the Gold league, and holding a #1 weekly ranking in Sri Lanka. Author of independent vulnerability research on SetupAPI callback redirection for EDR evasion, alongside developer of offensive security frameworks including Artemis and the Native Adversary Framework in C and x86-64 Assembly. Certified in Binary Fuzzing and Reversing (CBFRPro), Penetration Testing (CAPT), and Windows Offensive APIs. Seeking an offensive security or penetration testing internship to contribute to VAPT engagements, web and API security assessments, and vulnerability research while applying low-level tradecraft and adversary emulation skills.

## TECHNICAL SKILLS

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**Languages:** C/C++, x86-64 Assembly, Bash Scripting, SQL

**Offensive Security:** Reverse Engineering, Windows Internals, AV/EDR Internals, AV/EDR Evasion, Exploit Development, Offensive Tool Engineering, Malware Analysis, Red Teaming, Penetration Testing

**Security Assessment:** Vulnerability Assessment, OWASP Top 10, OSINT, Social Engineering, Privilege Escalation, Post-Exploitation, Enumeration, Active Directory, MITRE ATT&CK

**Security Tools:** Nmap, Metasploit Framework, MSFvenom, Burp Suite, OWASP ZAP, Wireshark, Wazuh, Splunk, SIEM, Nessus, OpenVAS, FFUF, Netcat, Scapy, Social Engineering Toolkit (SET)

**RE & Analysis Tools:** IDA Pro, Ghidra, x64dbg, CFF Explorer, Microsoft Sysinternals Suite

**Systems & Platforms:** Windows, Kali Linux, Linux (Debian), Virtualization, Docker, Git

## CERTIFICATIONS

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**Certified Binary Fuzzing & Reversing Professional (CBFRPro) – SecOps Group**

*Issued Jul 2026*

**Certified Associate Penetration Tester (CAPT) – Hackviser**

*Issued Feb 2026*

**Certified Windows Offensive API – Red Team Leaders**

*Issued 2026*

**Introduction to Linux LFS101 – The Linux Foundation**

*Issued 2026*

## COMPETITIVE SECURITY & TRAINING

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**TryHackMe – Practical Cybersecurity & Competitive Hacking**

Dec. 2024 – Present

*Hands-on Penetration Testing and Threat Hunting Platform*

- Ranked in the **Top 5%** globally; secured **1st place** in Silver and Bronze leagues, **11th place** in Gold league, and **#1 weekly ranking** in Sri Lanka
- Completed 142+ hands-on labs covering VAPT, web application security, API security testing, memory forensics, OS hardening, and vulnerability chaining
- Documented vulnerabilities, attack paths, and remediation recommendations across lab exercises; tools: Nmap, Burp Suite, OWASP ZAP, FFUF, Netcat

**Practical Low Level Exploitation & Reverse Engineering**

Apr. 2026 – Present

*PWN College*

- Advanced low-level exploitation and reverse engineering training in sandboxed environments
- Built custom shellcode and ROP chains; studied memory corruption, privilege escalation, and OS architecture

**Network Technician – NVQ Level 4**

Jan. 2025 – Present

*Technical College, Gampaha*

*Gampaha, Sri Lanka*

- Information and Communication Technologies; networking fundamentals and applied network security

## PROJECTS

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### **Artemis** | *C, x86-64 Assembly, Windows NT* 2026 – Present

- Low-level Windows NT offensive framework featuring a syscall resolver supporting 14,000+ Windows builds
- Implemented inline and indirect syscall stubs engineered for EDR telemetry evasion
- Integrated XOR, Rolling XOR, and RC4 payload encryption schemes
- [github.com/0x3xp/Artemis](https://github.com/0x3xp/Artemis)

### **Native Adversary Framework** | *C, x86-64 Assembly* 2026 – Present

- Windows process injection framework implementing process hollowing and thread hijacking techniques
- PEB walking and API hashing implemented entirely in x64 Assembly for stealth and portability
- Custom PE parsers for IAT/EAT analysis enabling dynamic import resolution without Windows API
- [github.com/0x3xp/Native-Adversary-Framework](https://github.com/0x3xp/Native-Adversary-Framework)

### **SENTINEX** | *C, Windows x64* 2026 – Present

- Stealth-focused, modular Windows reconnaissance framework designed for red team operations and adversary simulation
- Features a low-noise operator shell with modular recon commands, AV/EDR exposure awareness, and LOLBins discovery
- Covers Windows system, network, and security enumeration with minimal operational footprint
- [github.com/0x3xp/SENTINEX](https://github.com/0x3xp/SENTINEX)

## RESEARCH & VULNERABILITY DISCLOSURE

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### **Control Flow Redirection via SetupAPI Callback** | *Independent Security Research* 2026

- Documented methodology to subvert `setupapi.dll` callback routines using executable memory regions, bypassing endpoint detection telemetry that monitors conventional execution APIs
- Published a C proof-of-concept alongside defensive mitigation strategies
- [github.com/0x3xp/\[...\]/SetupAPI-Callback-Redirection](https://github.com/0x3xp/[...]/SetupAPI-Callback-Redirection)

### **WdFilter.sys Security Bypass** | *Responsible Disclosure – MSRC* 2026

- Identified a security bypass vulnerability in `WdFilter.sys`, the Windows Defender kernel-mode minifilter driver responsible for file system monitoring and real-time threat interception
- Vulnerability enables evasion of Defender's kernel-level detection pipeline; reported to Microsoft Security Response Center (MSRC) and currently under active review
- Full technical details remain confidential pending vendor remediation and coordinated public disclosure

## LANGUAGE SKILLS

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| Mother Tongue | English          |                |                |               |                   | CEFR Self-Assessment |
|---------------|------------------|----------------|----------------|---------------|-------------------|----------------------|
| Sinhala       | <i>Listening</i> | <i>Reading</i> | <i>Writing</i> | <i>Spoken</i> | <i>Production</i> | <i>Interaction</i>   |
|               | B2               | B2             | B1             | A2            |                   | A1                   |

## REFERENCES

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### **The Principal**

Technical College, Gampaha  
*Institutional referee confirming course enrollment, academic compliance, and performance standards*  
 Phone: (+94) 332237579

### **Malinda Fernando**

Lecturer, Technical College, Gampaha  
*Academic referee for practical technical skills, project development, and laboratory performance*  
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